COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 / Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:		**DATE SUBMITTED:**	
NAME:	Ace Tadalan	**INSTRUCTOR:**	Engr. Quejado

ACTIVITY NO. 8 **Install, Configure, and Manage Availability Monitoring tools**

1. PROCEDURE OUTPUT

What procedures did you perform?

1. Create a playbook that installs Nagios in both Ubuntu and CentOS. Apply the concept of creating roles.
2. Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)
3. Show an output of the installed Nagios for both Ubuntu and CentOS.
4. Make sure to create a new repository in GitHub for this activity

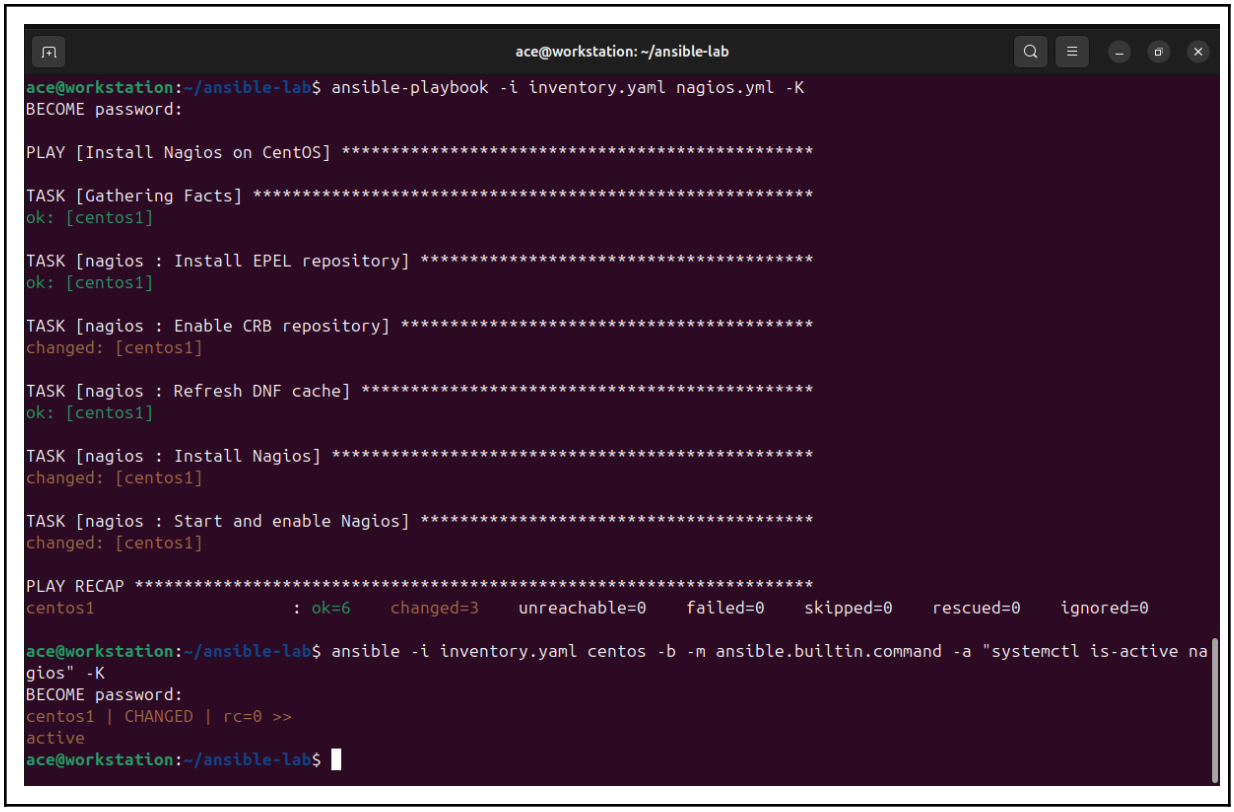
```
  GNU nano 7.2    roles/nagios/tasks/main.yml *
---
- name: Install EPEL repository
  ansible.builtin.yum:
    name: epel-release
    state: present
  become: yes

- name: Install Nagios
  ansible.builtin.yum:
    name:
      - nagios
      - nagios-plugins-all
    state: present
  become: yes

- name: Start and enable Nagios
  ansible.builtin.service:
    name: nagios
    state: started
    enabled: yes
  become: yes
```

```
  GNU nano 7.2              nagios.yml
---
- name: Install Nagios on CentOS
  hosts: centos
  become: yes

  roles:
    - nagios
```


ace@workstation: ~/ansible-lab
ace@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml nagios.yml -K
BECOME password:
PLAY [Install Nagios on CentOS] **
TASK [Gathering Facts] ***
ok: [centos1]
TASK [nagios : Install EPEL repository] **
ok: [centos1]
TASK [nagios : Enable CRB repository] **
changed: [centos1]
TASK [nagios : Refresh DNF cache] **
ok: [centos1]
TASK [nagios : Install Nagios] ***
changed: [centos1]
TASK [nagios : Start and enable Nagios] **
changed: [centos1]
PLAY RECAP ***
centos1 : ok=6 changed=3 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
ace@workstation:~/ansible-lab$ ansible -i inventory.yaml centos -b -m ansible.builtin.command -a "systemctl is-active nagios" -K
BECOME password:
centos1 | CHANGED | rc=0 >>
active
ace@workstation:~/ansible-lab$

```
ace@workstation:~/ansible-lab$ ansible -i inventory-nagios.
yaml workstation -m ansible.builtin.ping
localhost | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ace@workstation:~/ansible-lab$ ansible -i inventory-nagios.
yaml centos -m ansible.builtin.ping
centos1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
ace@workstation:~/ansible-lab$ ansible-playbook -i inventor
y-nagios.yaml nagios.yml -K
BECOME password:

PLAY [Install Nagios on CentOS] ****************************
**********************

TASK [Gathering Facts] *************************************
**********************
ok: [centos1]

TASK [nagios : Install EPEL repository] ********************
**********************
ok: [centos1]
```

```
PLAY RECAP *********************************************************
**********************
centos1                    : ok=6    changed=1    unreachab
le=0    failed=0    skipped=2    rescued=0    ignored=0
localhost                  : ok=3    changed=2    unreachab
le=0    failed=0    skipped=5    rescued=0    ignored=0

ace@workstation:~/ansible-lab$
```

```
ace@workstation:~/ansible-lab$ systemctl is-active nagios4
active
ace@workstation:~/ansible-lab$ systemctl status nagios4 --n
o-pager
● nagios4.service - nagios4
     Loaded: loaded (/usr/lib/systemd/system/nagios4.servic
e; enabled; preset: enabled)
     Active: active (running) since Sat 2026-09-19 23:21:51
 PST; 2min 9s ago
       Docs: man:nagios4
   Main PID: 17575 (nagios4)
      Tasks: 6 (limit: 4417)
     Memory: 2.6M (peak: 3.4M)
        CPU: 81ms
```

```
ace@workstation:~/ansible-lab$ ansible -i inventory-nagios.
yaml centos -b -m ansible.builtin.command -a "systemctl is-
active nagios" -K
BECOME password:
centos1 | CHANGED | rc=0 >>
active
ace@workstation:~/ansible-lab$ ansible -i inventory-nagios.
yaml centos -b -m ansible.builtin.command -a "systemctl sta
tus nagios --no-pager" -K
BECOME password:
centos1 | CHANGED | rc=0 >>
● nagios.service - Nagios Core 4.4.14
     Loaded: loaded (/usr/lib/systemd/system/nagios.service
; enabled; preset: disabled)
     Active: active (running) since Sat 2026-09-19 23:09:31
 PST; 15min ago
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the folowing:
1. What are the benefits of having an availability monitoring tool?

Availability monitoring tools help with efficiency and to be able to continuously check whether servers and services are running properly. As they can detect whether it is downtime or service problems quickly, allowing issues to be addressed before they significantly affect users quivkly.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of

concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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